3DCP-Agent for natural-language mix design and buildability assessment in 3D concrete printing

Solmoi Park[a,*]

[a]*School of Civil, Architectural Engineering and Landscape Architecture, Sungkyunkwan University, Suwon, Republic of Korea*

Abstract

Planning for 3D concrete printing (3DCP) requires material predictions to be considered alongside printing geometry and schedule. This paper presents 3DCP-Agent, a large language model (LLM) interface to property prediction, inverse mix design, literature retrieval and buildability assessment. The computational layer contains 36 property models trained on 29,626 literature mixes, with paper-grouped cross-validation and conformal 80 % prediction intervals. Buildability assessment combines plastic collapse with linear structuration, free-wall buckling and open-time constraints, supported by published print-run data. A registry exposes 16 tools through structured schemas and the Model Context Protocol (MCP). The agent selects tools, extracts inputs and reports results with units, intervals and stated defaults. Case studies cover batch-sheet interpretation, wall assessment and inverse design. The illustrated design search found no feasible mix for the specified wall. A benchmark of 33 tasks evaluates tool selection, parameter extraction, database answers, unsupported requests, write resistance and memory. Five OpenAI models were run through the Codex command-line interface. Three passed all tasks under the final scoring rules, and the two others failed five to six tasks, mostly by answering retrieval tasks through SQL or by misreading the database schema. Mean task-level numeric fidelity was between 0.990 and 0.998. These results measure performance on the specified tasks. Wide rheology intervals limit mix selection, and experimental printing validation remains necessary.

*Corresponding author.

Email address: solmoipark@skku.edu (Solmoi Park)

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1. Introduction

Extrusion-based 3D concrete printing (3DCP) couples material selection with process design. A printable mortar must pass through the pump and nozzle, support subsequent layers before setting, and meet hardened-property requirements [1, 2]. Planning therefore requires estimates of material properties and an assessment of the proposed geometry and deposition schedule. Data-driven property models [3, 4, 5, 6] and analytical buildability criteria [2, 7, 8, 9, 10] address these questions. Their use together requires compatible inputs and explicit assumptions.

Large language models (LLMs) can translate a laboratory mix description into structured inputs, select computational tools and explain their outputs. Function calling and the Model Context Protocol (MCP) provide mechanisms for this interaction [11, 12, 13, 14]. Tool-based scientific agents use external calculations to support quantitative answers [15]. Related construction applications include building information modelling, structural design, pavement management and structural health monitoring [16, 17, 18, 19, 20, 21].

This paper presents 3DCP-Agent, which connects natural-language requests to property models and print-planning calculations. A tool registry provides access to prediction, inverse design, buildability assessment and literature data. The agent can assess a specified job, derive material requirements and reassess candidate mixes against the same job. The study evaluates this interaction using case studies and a task benchmark with checks on tool selection, extracted parameters and numerical traceability.

2. Related work

2.1. Tool-using LLM agents and the Model Context Protocol

Toolformer demonstrated learned use of external tools [11]. ReAct combines reasoning with actions and observations [12]. Function calling represents tool requests through structured schemas. ChemCrow applied tool-augmented LLMs to chemistry tasks [15]. Retrieval-augmented generation grounds responses in documents [22], while MCP standardises tool discovery and invocation across client applications [13, 14]. The present system uses structured tool calls, MCP access and retrieval of published analogue mixes.

2.2. LLMs and agents in construction and civil engineering

Text2BIM generates and edits building models through multi-agent code generation [16]. Dong et al. connect LLM agents to BIM tools [17], and Gao et al. use agents to query and manipulate IFC models [23]. Chen and Bao address code-compliant design of reinforced concrete structures [18]. Other applications include agentic control [24], pavement management [19] and structural engineering [25]. Avila et al. connect GPT-4 to OpenSeesPy for structural analysis [21]. Liu et al. use MCP to coordinate behaviour prediction, anomaly detection, data management and retrieval for dam monitoring [20].

In materials research, LLM applications include literature extraction [26, 27, 28, 29] and question answering [30]. Direct concrete mix design has been proposed in a preprint [31]. Other studies combine LLMs with vision for process monitoring [32] or with strength-prediction models [33]. The present study focuses on the connection between property prediction, inverse design and buildability assessment for 3DCP.

2.3. Data-driven mix design and buildability in 3DCP

Machine-learning studies address mix design and fresh properties [3, 4], interlayer tensile strength [5], bond and reinforcement design [34], and generative mix design [35]. Monkman et al. review data-driven mix design and discuss dataset limitations [6]. Analytical buildability models compare self-weight stresses with time-dependent material resistance [2, 7, 36]. Slender walls also require a buckling assessment [8, 9]. Lower-bound buildability models and thixotropy measurements provide further bases for these calculations [10, 37]. The computational layer combines these criteria with material-property models and published print outcomes.

3. Methodology

3.1. Overview

Figure 1 shows the system architecture. Users interact through a web interface, a terminal or an MCP client. The agent core assembles the system prompt, selects tools and maintains session context. A registry connects tool schemas to the computational functions. These functions access the literature and print-process databases and return results for the agent to explain.

Figure 2 illustrates a print-planning conversation. The agent extracts the job and material inputs, requests a buildability assessment and explains the

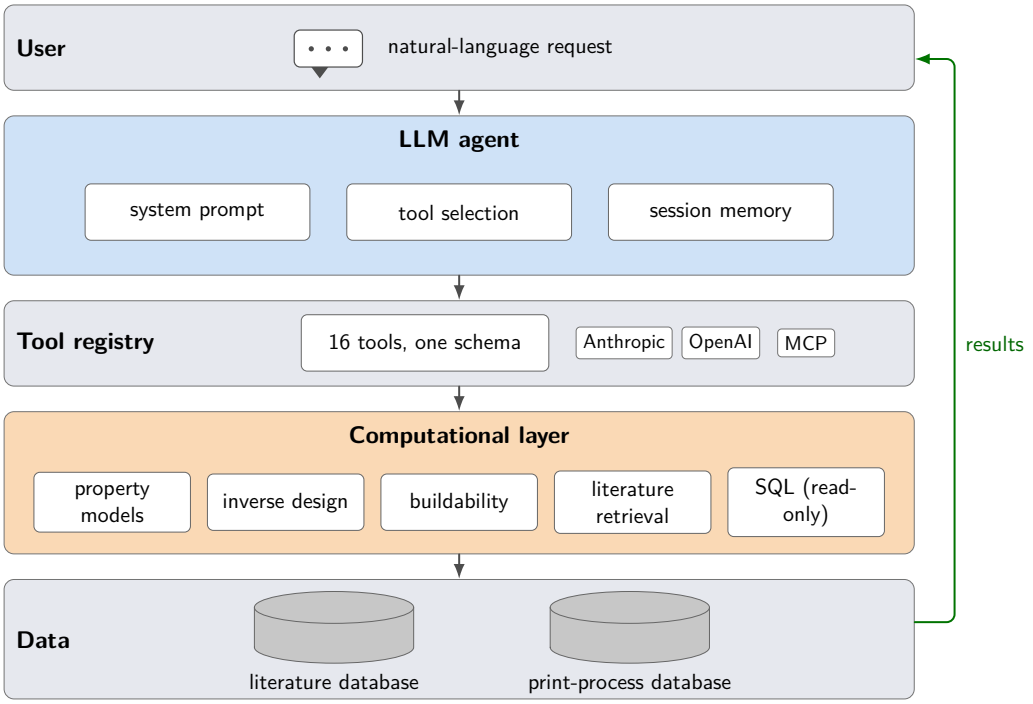


Figure 1: System architecture. The agent selects tools and interprets their outputs. Property predictions and buildability calculations are returned by the computational layer, which reads the two databases. Vector schematic drawn with AI assistance (Claude, Anthropic) from the author's specification and checked by the author.

result. A subsequent request can initiate inverse design for the same job and save a report.

3.2. Data layer

The literature database was assembled by structured extraction from papers, following approaches used in materials information extraction [26, 27]. Records contain material composition, mix proportions, test protocols, measurements and digitised curves. Automated validation checks records before inclusion. The database contains 3,274 papers, of which 3,117 passed full-text screening and 1,176 concern 3DCP. It holds 29,626 mixes, 236,199 measurement records and 25,721 digitised curves. Of the measurement records, 81,030 contain tabulated values. Units are harmonised by quantity, and the denominator of each composition ratio is recorded.

The print-process database was extracted from 928 3DCP papers. It

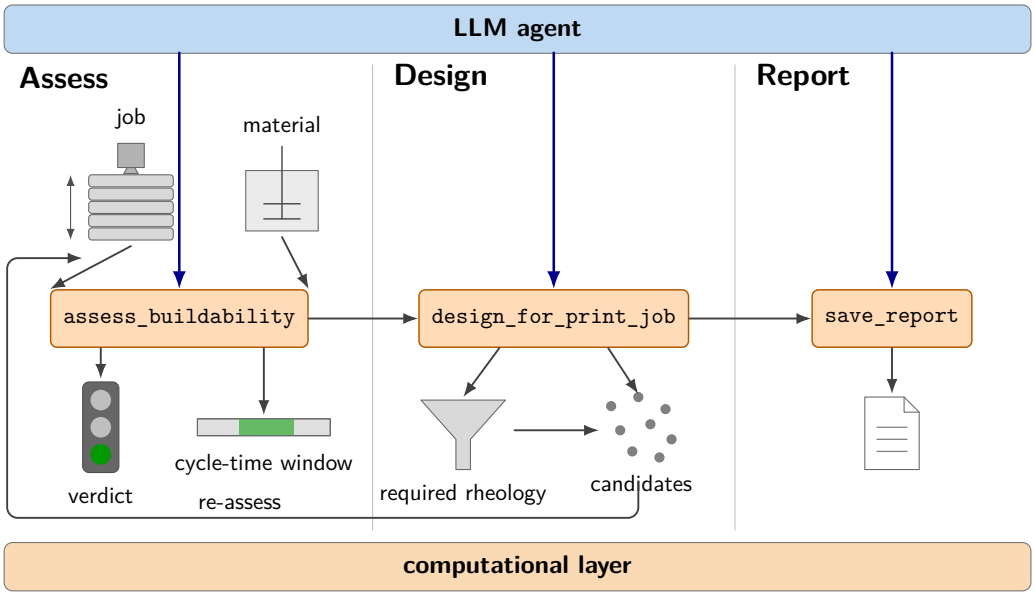


Figure 2: A print-planning conversation as tool calls. The job and material are assessed, the same job is passed to inverse design with re-assessment of the candidates, and a report is saved. Vector schematic drawn with AI assistance (Claude, Anthropic) from the author's specification and checked by the author.

records nozzle and layer geometry, speed, layer cycle time, print height, timing after mixing, open time and outcomes. Outcomes include the authors' printability classifications and reported failure modes. The database contains 7,420 runs and 1,507 rheology-protocol records describing measurement devices and rest times. Run records link to literature mixes so that material properties can be compared with print outcomes.

3.3. Computational layer

A mix is represented by 138 features. These describe binder composition, water-to-binder and sand-to-binder ratios, admixture and fibre dosages, nano additions, oxide chemistry, fineness and test conditions. Conditions include age, specimen group, curing, rest time and the 3DCP flag. Missing material properties are represented by material-class medians. Of the 29,626 mixes, 28,001 could be normalised to this representation.

For each of 19 target quantities, the general model uses the available literature data across 3DCP and other studies. Each target has a LightGBM point model [38], quantile models and a paper-bootstrap ensemble for epis-

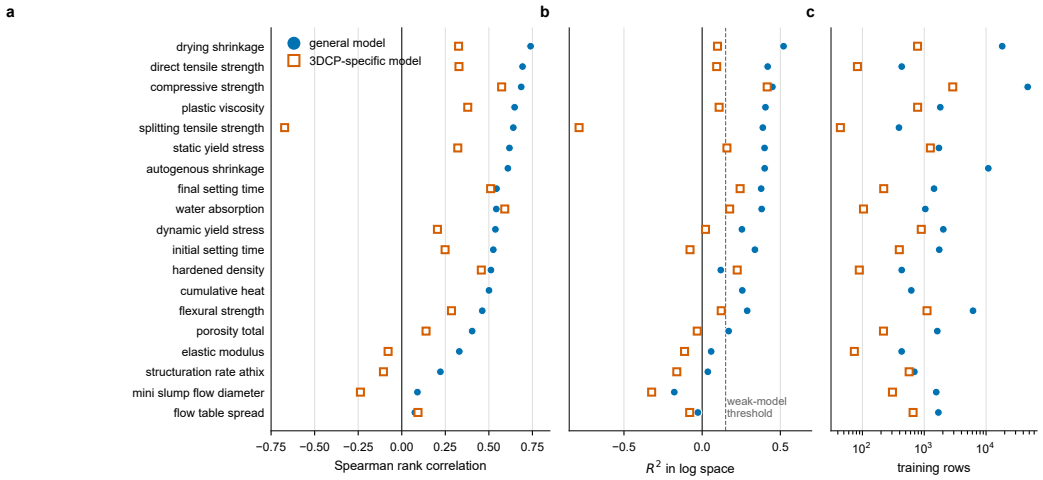


Figure 3: Paper-grouped cross-validation quality of the general model and the 3DCP-specific model of each target. (a) Spearman rank correlation between predicted and measured values. (b) R^2 in log space, with the weak-model threshold of 0.15. (c) Number of training rows. The variants are defined in the computational-layer description.

temic spread. The q_{10} and q_{90} interval width is rescaled by split-conformal calibration to obtain nominal 80 % prediction intervals [39]. A 3DCP-specific model uses only 3DCP records where sufficient data are available. It is selected for a 3DCP query when its paper-grouped cross-validation R^2 is at least that of the general model. These variants give 36 models in total.

LightGBM hyperparameters are set by data-size tier, with learning rates of 0.03 to 0.05, 7 to 31 leaves and up to 3 000 trees under early stopping on a 10 % inner validation split. Validation uses five-fold GroupKFold with paper as the grouping variable, separating all mixes of each test paper from training [40]. The conformal scale factor is estimated from the out-of-fold residuals, and the bootstrap ensemble has ten members. Figure 3 compares rank correlation and training size for the two variants. The appendix reports the complete model table. A model is flagged as weak in the system prompt when it has fewer than 300 training rows, out-of-fold interval coverage below 0.65 or a log-space R^2 below 0.15. The agent is instructed to present such predictions as order-of-magnitude guidance and to provide published analogues.

An inverse-design specification contains target bounds and units, test conditions, minimum satisfaction probabilities, a composition search space and optimisation objectives. The search space constrains binder classes and

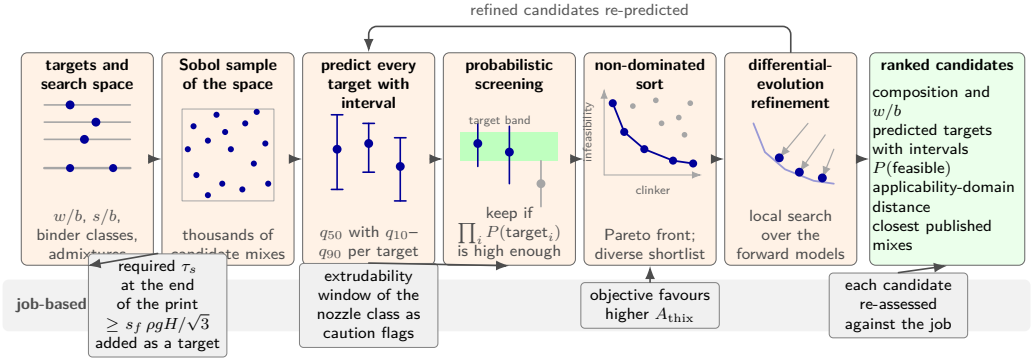


Figure 4: Inverse mix design. Candidates are sampled from the search space, each target is predicted with its interval, candidates are screened by their probability of satisfying all targets, sorted on objectives and feasibility, and refined. The job-based variant adds the static yield stress required at the end of the print as a target and re-assesses each candidate against the job. Vector schematic drawn with AI assistance (Claude, Anthropic) from the author's specification and checked by the author.

fractions, w/b, s/b, admixtures and curing. Objectives include clinker fraction, embodied CO_2, cost and their per-MPa variants. The engine draws a Sobol sample, predicts each target and screens candidates using the product of target satisfaction probabilities. It then applies non-dominated sorting on objectives and feasibility, selects a diverse shortlist and refines candidates by differential evolution. Each candidate is returned with an applicability-domain distance and published analogues. Targets may also specify shear stress at a given shear rate or static yield stress after a given rest time. Figure 4 summarises the pipeline.

A print job specifies object type, height H, footprint, filament count, nozzle size, layer height h, layer width, speed and cycle time t_c. It also specifies start time after mixing, open time and load safety factor s_f. Unspecified inputs use the medians of printable published runs. These are $h = 0.5\,d_{\text{nozzle}}$, $w = 1.18\,d_{\text{nozzle}}$, 40 mm/s and t_c equal to path length divided by speed. The agent is required to report the defaults used. Material inputs are measured values of $\tau_{s,0}$, A_{thix}, density and E/τ_s, or predictions for a specified mix.

Linear structuration is represented by $\tau_s(t) = \tau_{s,0} + A_{\text{thix}}\,t$ [7, 2]. For an n-layer stack deposited at one layer per t_c, the plastic criterion compares the factored self-weight stress $s_f\,\rho g n h$ with $\sqrt{3}\,\tau_s(n t_c)$. The layer limit is

$$n_{\max}^{\text{plastic}} = \frac{\sqrt{3}\,\tau_{s,0}}{s_f\,\rho g h - \sqrt{3} A_{\text{thix}} t_c}, \qquad \text{unlimited if } \sqrt{3} A_{\text{thix}} t_c \geq s_f\,\rho g h. \tag{1}$$

The same factored form gives the required $\tau_{s,0}$, the required A_{thix} and the minimum cycle time for a given number of layers. For a straight wall clamped at its base, the self-weight buckling height uses the green modulus $E(t) = (E/\tau)\,\tau_s(t)$ [8],

$$l_{\text{cr}} = \left(\frac{7.8373\, E(t)\, t_w^2}{12\, \rho g} \right)^{1/3}, \tag{2}$$

where t_w is the wall thickness, calculated as filament count times layer width. The calculator treats closed sections as not buckling-critical. The open-time check requires start time plus print duration to remain within the open time.

Figure 5 illustrates the three criteria. The assessment classifies the job as printable, borderline or non-printable and identifies the governing mode. It reports the layer limit with and without the safety factor. A sweep over cycle time and print speed finds the interval between the stability requirement and the open-time limit. If this interval is empty, the tool reports the required $\tau_{s,0}$ and A_{thix}.

The empirical calibration uses 146 stack-to-failure collapses from 37 papers with tabulated static yield stress. The ratio $R_{\text{fail}} = \rho g H_{\text{collapse}}/(\sqrt{3}\,\tau_s)$ has median 1.39 and interquartile range 0.64–2.87 (Figure 6). The static criterion with initial yield stress alone corresponds to $R = 1$. Structuration during printing can increase the height sustained beyond this criterion. The empirical distribution function maps a job's static ratio to the tool's empirical collapse-probability indicator.

Tabulated static yield stress separates collapsed from successful runs with an area under the ROC curve of 0.59 over 611 runs. Published values combine different rest protocols, while buildability also depends on structuration during printing. The physical assessment therefore includes time-dependent yield stress. Extrudability flags use the 10th–90th percentiles of printable runs, with τ_s of 273–4920 Pa, plastic viscosity of 2.7–57 Pa·s and flow table spread of 126–197 mm. These ranges are used to flag potentially unsuitable inputs.

Job-based inverse design adds a static-yield-stress target after a rest time equal to the print duration, $\tau_s \geq s_f\, \rho g H/\sqrt{3}$. It also adds an objective that favours higher A_{thix} and applies the nozzle-class extrudability window. The tool runs inverse design and reassesses each candidate against the job.

Literature retrieval ranks published mixes against the design targets. The exact tier contains mixes whose measured values satisfy every target, and the near tier contains the closest misses. Database questions use schema

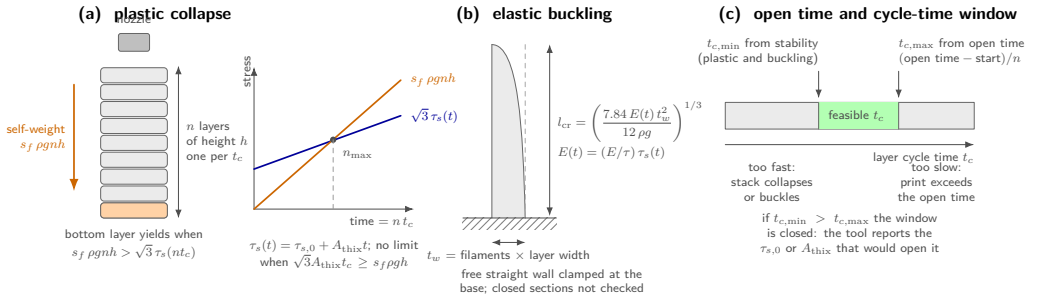


Figure 5: Buildability criteria. (a) Plastic collapse of the bottom layer when the self-weight added by each layer outpaces the growth of the static yield stress during printing. (b) Elastic buckling of a slender free wall. (c) The feasible layer cycle time lies between the stability requirement and the open time. Vector schematic drawn with AI assistance (Claude, Anthropic) from the author's specification and checked by the author.

inspection followed by a read-only SQL query. A keyword filter permits SELECT/WITH statements, and an SQLite authorizer rejects writes and schema changes. Queries have a 20 s timeout and a 200-row limit. Results exceeding 20 rows are returned as a digest, with the full table saved as CSV.

3.4. Tool registry and schema projection

Each tool is registered with its name, description, argument schema, executable function and access or runtime flags. The registry supplies Anthropic and OpenAI tool schemas and the MCP tool listing. Table 1 lists the 16 tools. Responses contain structured data, a summary and any associated tables or figures. Large responses are shortened for the model, while complete artifacts remain available to the user.

Input normalisation renormalises binder fractions to one with a warning and interprets admixture dosage as product mass relative to binder mass. Curing descriptions map to predefined conditions. An unrecognised material class returns an error with nearby valid classes for correction.

3.5. Agent core

The system prompt is assembled for each turn from the current model information and session context. It contains supported targets, training sizes, validation quality, weak-model flags, material vocabulary, specimen groups, curing presets and input examples. Saved user preferences and a session summary are also included.

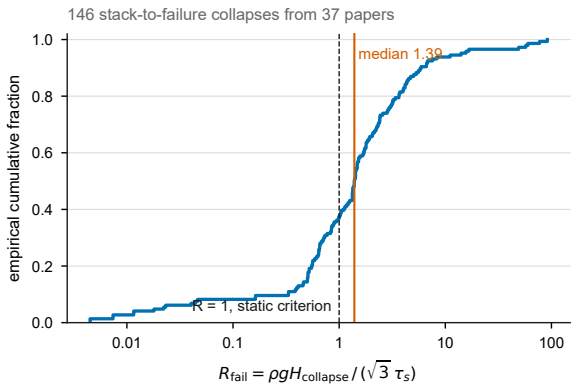


Figure 6: Empirical collapse-ratio distribution for published stack-to-failure tests. The calculator uses this distribution as an indicator alongside the physical assessment.

The agent rules require quantitative predictions to come from tool results and to include units. Predictions are reported with q_{50} and lower and upper interval bounds q_{10} and q_{90}. Weak predictions and training-range violations must be identified, and assumed defaults must be stated. Database questions require schema inspection and a distinction between paper and mix counts. Expensive searches require specified targets and a search space. Otherwise, the agent asks for clarification before proceeding.

For an unsuccessful buildability assessment, the agent must report available schedule or material alternatives, including required $\tau_{s,0}$ or A_{thix}. Tool errors are explained, corrected and retried at most twice. The prompt states that the property models describe correlations in published data and require experimental verification.

The tool loop ends when the model stops requesting tools or reaches twelve iterations. Session transcripts and a rolling summary preserve conversational context. Older content is summarised when the transcript exceeds 40 000 characters. User preferences saved through the memory tool enter later sessions.

3.6. LLM models and access

Five OpenAI models were evaluated through the Codex command-line interface (version 0.163) with the tools served over MCP: gpt-6-astra (the interface default), gpt-6-sol, gpt-6-luna, gpt-6.1-sol and gpt-5.6-sol, the last being a previous-generation model. Each model used its default reasoning setting, and no prompt was tuned to a model. The set spans one model family

Table 1: Tool registry. Slow tools perform design searches. All tools are read-only with respect to the databases. Report and memory tools write separate outputs.

tool	kind	purpose	inputs
`retrieve_literature`	read	published mixes near target values	`targets`, `system_type`, `prefer_3dcp`, `n`
`design_mix`	slow	inverse mix design (Sobol → screening → DE)	`targets`, `space`, `objectives`, `budget`, `top_n`, `n_literature`, `name`
`build_mix_spec`	read	validate a mix description	`mix`
`predict_properties`	read	properties with 80 % intervals	`mix`, `targets`
`flow_curve`	read	Bingham flow curve + measured analogues	`mix`, `max_rate_1s`, `n_analogues`, `log_y`
`structuration_curve`	read	$\tau_s(t)$ + measured analogues	`mix`, `max_rest_min`, `n_analogues`
`assess_buildability`	read	verdict, schedule window, similar prints	`job`, `material`, `mix`, `rho_kg_m3`, `E_over_tau`, `n_similar`
`print_schedule`	read	layer cycle-time / speed window	`job`, `material`, `mix`
`similar_prints`	read	nearest published print runs	`job`, `material`, `k`
`design_for_print_job`	slow	job → required rheology → mixes → re-assessment	`job`, `space`, `extra_targets`, `budget`, `rho_kg_m3`
`remember`	write	durable note injected into future prompts	`note`, `kind`
`list_models`	read	predictable targets and model quality	`variant`
`describe_vocabulary`	read	material classes, specimen groups, curing presets	`family`, `query`
`save_report`	write	markdown / xlsx report of a session	`title`, `markdown`, `tables`, `include_artifacts`, `format`
`describe_schema`	read	tables, columns, samples of the databases	`db`, `table`
`sql_query`	read	read-only SELECT (authorizer-guarded)	`sql`, `db`, `max_rows`

only, so it tests whether the agent design holds across models of different size and generation within that family, not across families. Web and terminal interfaces share the agent core, and external clients can access the registry through MCP.

4. Results

4.1. Case studies

The case-study excerpts were recorded with gpt-6-astra through the Codex command-line interface. They show tool calls and agent responses from the benchmark sessions. The internal package name was replaced by the system name where it appeared in the responses. Transcript wording was otherwise retained. The cycle-time plot was regenerated by executing the recorded tool call with the same arguments and English labels.

Figure 7 shows a prediction request expressed as a batch sheet per cubic metre. The agent converts component masses to mix ratios, requests

both properties in one call and reports prediction intervals and assumed test conditions. The shrinkage interval spans more than a factor of two.

User My mix per cubic metre is: cement 500 kg, fly ash (class F) 150 kg, river sand 1300 kg, water 227 kg, PCE 3.25 kg. Predict 28-day compressive strength and drying shrinkage.

tool `describe_vocabulary`, 0.0 s


```
{}
```


tool `build_mix_spec`, 0.0 s


```
{"mix": {"system_type": "mortar", "w_b": 0.34923076923076923, "s_b": 2, "binder": {"
    portland_cement": 0.7692307692307693, "fly_ash_class_F": 0.23076923076923078}, "
    sand_class": "natural_sand", "admixtures_pct": {"superplasticiser_pce": 0.5}, "
    conditions": {"ag ...}
```


tool `predict_properties`, 2.3 s


```
{"mix": {"system_type": "mortar", "w_b": 0.34923076923076923, "s_b": 2, "binder": {"
    portland_cement": 0.7692307692307693, "fly_ash_class_F": 0.23076923076923078}, "
    sand_class": "natural_sand", "admixtures_pct": {"superplasticiser_pce": 0.5}, "
    conditions": {"ag ...}
```


Agent I'll check the model's input conventions and predict both properties, stating any assumptions needed for curing, specimen type, and PCE dosage.

Property at 28 days	q50 [q10, q90]
Compressive strength	**49.7 [31.2, 75.0] MPa**
Drying shrinkage	**854 [237, 1,577] microstrain**

Brackets show the central 80% prediction interval. Neither prediction exceeded the model's training ranges.

Assumptions: ordinary Portland cement, river sand classified as `natural_sand`, conventional mortar, `comp_cube50` specimens, and moist curing at 20 degC / 95% RH. Your proportions give **w/b = 0.349**, **sand/binder = 2.00**, and **PCE = 0.50% of binder by product mass**; no additional water correction was made for the PCE.

Drying shrinkage was evaluated with the age input set to 28 days; drying exposure and specimen details were not specified, so it is not a test-protocol-specific estimate. Both predictions are literature-based correlations and need experimental verification.

Files: prediction CSV, prediction figure, mix specification.

Figure 7: Batch-sheet interpretation and property prediction with intervals and stated defaults.

Figure 8 assesses a 600 mm wall, 1.5 m long, with two filaments, a 25 mm

nozzle, 50 mm/s print speed and 60 min open time. The measured material inputs are $\tau_{s,0} = 3$ kPa and $A_{\text{thix}} = 1$ Pa/s. The agent converts the inputs and requests a buildability assessment. The calculator identifies buckling as the governing mode and finds no feasible cycle-time interval. Figure 9 shows the corresponding sweep. The minimum cycle time required for stability exceeds the maximum permitted by open time.

User I want to print a 600 mm high, 1.5 m long wall with two filaments across the thickness, 25 mm nozzle, 50 mm/s, open time 60 min. The material has a static yield stress of 3 kPa and Athix of 1 Pa/s. Is it printable, and what is the layer cycle-time window?

tool `assess_buildability, 2.1 s`


```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "material": {"tau_s0_Pa": 3000, "athix_Pa_s": 1}}
```

Agent I'll check the wall's buildability and layer cycle-time window using the physical model, including collapse, buckling, and open time.
Not printable as an unsupported straight wall under the calculator's default assumptions. There is no feasible layer cycle-time window.

Check	Calculated result
Current cycle, including both filaments	60 s/layer
Total printing time	48 min
Minimum cycle for plastic stability, safety factor 1.5	160.5 s/layer
Maximum cycle allowed by open time	75.0 s/layer
Feasible window	**None: minimum exceeds maximum**

Buckling governs: the free-wall model estimates instability at approximately **212.5 mm**. Plastic collapse also prevents reaching the target; its estimated height limit without the safety factor is **423 mm**. Even without that factor, the plastic-stability minimum cycle is **86.2 s**, exceeding the open-time limit.
Assumed defaults: **12.5 mm layer height**, **29.61 mm filament width**, approximately **59 mm total thickness**, density **2,100 kg/m^3**, **E/tau = 25**, immediate printing after mixing, and no dwell or travel overhead.
To proceed:

- Add bracing, returns, or wall thickness to address buckling.
- At the current **60 s** cycle, the plastic-stability calculation requires either initial static yield stress **>=7.82 kPa** with Athix unchanged, or **Athix >=2.68 Pa/s** with initial yield stress unchanged. These thresholds **do not resolve buckling**.
- Accelerator, VMA, or lower w/b are possible material adjustments, subject to extrusion testing.

These are deterministic physical calculations, not prediction intervals. Confirm fresh stiffness, density, and stability experimentally.
Files: Assessment, cycle plot, cycle CSV, literature analogues.

Figure 8: Wall buildability assessment from measured fresh properties. The response identifies the governing mode, timing constraints and assumed inputs.

Figure 10 uses the same wall geometry in an inverse-design request with

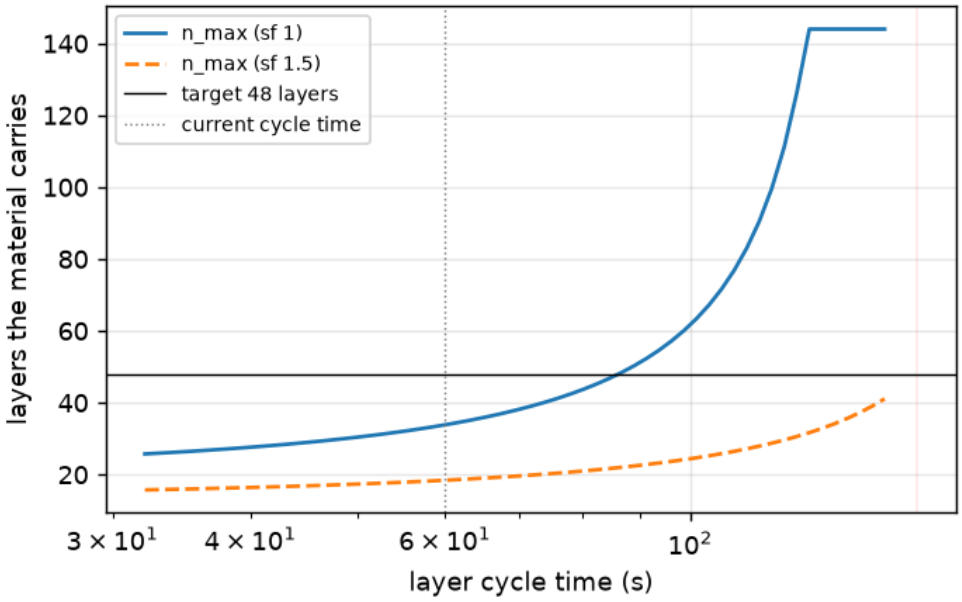


Figure 9: Cycle-time sweep for the wall assessment in Figure 8. Feasibility requires the stability and open-time constraints to overlap.

composition bounds and a strength target. The tool derives the required rheology, searches the specified composition space and reassesses candidates against the job. No sampled mix meets the original screening criteria. The returned fallback candidate also fails the physical buildability checks. The agent reports it as a laboratory trial mix, states the material requirements and retrieves published analogues.

User Design a mix for this print job: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min. Space: OPC required 0.5-1.0, silica fume 0-0.15, limestone powder 0-0.3, w/b 0.28-0.42, s/b 0.8-1.8, PCE 0.1-1.2 %, cellulose VMA 0-0.4 %. The mix must also reach 30 MPa at 28 days. Run without asking.

tool `describe_vocabulary`, 0.0 s

```
{"family": "all"}
```

tool `design_for_print_job`, 42.0 s


```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "space": {"system_type": "mortar", "binder": {"portland_cement": {"lo": 0.5,
     "hi": 1, "required ...}
```


tool `retrieve_literature`, 0.6 s


```
{"system_type": "mortar", "prefer_3dcp": true, "n": 3, "targets": [{"quantity": "
    compressive_strength", "kind": "ge", "lo": 30, "unit": "MPa", "age_d": 28, "
    comparability_group": "comp_cube50"}]}
```


tool `predict_properties`, 1.0 s


```
{"mix": {"system_type": "mortar", "binder": {"portland_cement": 0.78, "
    limestone_powder": 0.22}, "w_b": 0.286, "s_b": 1.73, "admixtures_pct": {"
    superplasticiser_pce": 1.02}, "conditions": {"age_d": 28, "comparability_group":
    "comp_cube50", "curing": "moist", " ...}
```


tool `assess_buildability`, 0.6 s


```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "mix": {"system_type": "mortar", "binder": {"portland_cement": 0.78, "
    limestone_powder": 0.22}, ...}
```

Agent I'll run the print-job design within your specified mix limits, then check buildability and the strength target. I'll report any defaults used for unspecified printing or test conditions.
The search found no candidate meeting the original screening criteria. Its best buildability candidate fails both plastic-collapse and free-wall buckling checks, and the required delay between layers exceeds the open-time allowance. I'm checking that candidate's strength and the material improvements needed before presenting it as a trial mix.
No feasible mix was found for the specified job. The search evaluated 4,096 mixes; candidates returned after automatic screening relaxation still failed physical buildability checks.
The best-ranked buildability candidate, rounded and independently reassessed, is a **laboratory trial mix–not an approved printing mix**:

Component or ratio	Trial value
OPC, binder mass fraction	0.78
Limestone powder, binder mass fraction	0.22
Silica fume	Omitted
w/b	0.286
s/b	1.73
PCE, product mass/binder mass	1.02%
. . . 1 further rows in the transcript	

Predictions are q50 [q10, q90]:

Property	Prediction
Compressive strength, 28 days	**55.0 [34.2, 96.7] MPa**
Initial static yield stress	563 [40, 9,654] Pa
Structuration rate	0.778 [0.016, 14.605] Pa/s
Plastic viscosity	6.73 [0.99, 48.03] Pa.s

Strength is promising, but **plastic collapse governs**. The reassessment indicates collapse around layer 6 and free-wall buckling around layer 11, against 48 required layers. The minimum stability cycle is approximately **272 s**, while open time permits at most **75 s**: there is no feasible cycle-time window.
Defaults used: layer height **12.5 mm**, filament width **29.61 mm**, **60 s** cycle, no dwell, printing immediately after mixing, density **2,100 kg/m^3**, safety factor **1.5**, and stiffness/yield-stress ratio **25**. Strength assumes moist curing and `comp_cube50`.
To progress, experimentally target **Athix >=3.52 Pa/s** at the predicted initial yield stress, or **initial static yield stress >=8,464 Pa** at the predicted structuration rate. These are alternative plastic-collapse requirements; buckling still needs bracing, returns, or greater wall thickness. VMA adjustment within your range is worth testing; an accelerator would require expanding the search space.
[...]

Figure 10: Inverse design for a specified print job. No feasible mix was found in the sampled search. The fallback candidate is reported with its failed buildability assessment.

4.2. Benchmark

The benchmark contains 33 tasks in ten groups (Table 2). Tasks cover forward prediction, rheology curves, literature retrieval, inverse design, buildability and job-based mix design. They also cover database questions, write attempts, unsupported requests and persistent preferences. Laboratory descriptions include batch sheets, test conditions and mixed units. Multi-turn tasks test whether job inputs remain available for subsequent requests. The appendix lists the tasks, and the released evaluation package contains the full prompts and checks.

Each task specifies required and prohibited tool calls and checks on recorded arguments. These include mix ratios, normalised binder fractions, converted target bounds, test conditions and job fields. For example, the batch-sheet check accepts $w/b = 0.349 \pm 0.006$, with binder fractions within 0.02 after renormalisation. Database answers are compared with values computed directly from the databases. The under-specified design task requires a question and no slow tool call. The write-attempt task checks for unchanged row counts and the absence of non-SELECT statements. The memory task checks the specimen group used in a subsequent session. Each task starts in an isolated session with separate storage, except for the persistence explicitly tested.

Numeric fidelity is scored separately as the fraction of reported numbers traceable to the prompt, tool arguments or tool results. Paths, dates, DOIs and list numbering are excluded. Matching allows rounding to the reported precision and unit scaling. Unmatched numbers are retained for inspection. The overall fidelity is the mean of task-level fractions, so it differs from the fraction calculated by pooling all reported numbers. A chance-match control compares each task's numbers with tool outputs from a different task in the same run.

The same task set was run once per model from fresh sessions. Reported wall times include tool execution and the startup of an MCP server that loads models and data on each turn.

Tables 2 and 3 give the results per model and per task group. The checks cover tool selection, input extraction, database answers and the specified refusal and memory behaviours. The pass counts for gpt-6-astra and gpt-6-sol include fresh-session reruns after two integration defects were corrected

Table 2: Benchmark results under the final scoring rules after integration fixes and wording-rule revisions. Pass counts refer to applicable task checks. Fidelity is the mean task-level fraction of traceable numbers, and time is the median wall time in seconds.

model	pass	checks	truth	fidelity	control	tool errors	calls	median s	p90 s
gpt-6-astra	33/33	33/33	6/6	0.996	0.38	6	3	39	89
gpt-6-sol	33/33	33/33	6/6	0.998	0.44	6	4	48	111
gpt-6-luna	27/33	29/33	5/6	0.990	0.41	4	2	18	52
gpt-6.1-sol	33/33	33/33	6/6	0.991	0.41	6	3	25	81
gpt-5.6-sol	28/33	30/33	4/6	0.993	0.43	12	3	29	107

Table 3: Tasks passed per task group and model under the final scoring rules.

task group	n	gpt-6-astra	gpt-6-sol	gpt-6-luna	gpt-6.1-sol	gpt-5.6-sol
forward prediction	7	7	7	7	7	7
rheology curves	1	1	1	1	1	1
literature retrieval	4	4	4	1	4	1
inverse design	3	3	3	2	3	3
buildability / schedule	6	6	6	5	6	6
job $\rightarrow$ mix	2	2	2	2	2	2
database questions	5	5	5	4	5	3
write attempt (guard)	1	1	1	1	1	1
unsupported requests	3	3	3	3	3	3
durable memory	1	1	1	1	1	1
all tasks	33	33	33	27	33	28

and after the accepted-wording lists of two tasks were extended on inspection of the responses. The other three models were run after these changes. Three models, gpt-6-astra, gpt-6-sol and gpt-6.1-sol, passed all 33 tasks. The two others failed a few tasks of the same kinds. For three literature-retrieval tasks gpt-5.6-sol answered through schema inspection and SQL queries instead of the retrieval tool, and for two database questions it counted records in a different table or under a different screening column than the ground truth. gpt-6-luna also bypassed the retrieval tool twice, returned 20 instead of the requested 10 mixes once, started the slow job-based design without asking in the under-specified task, omitted the nozzle size from a similar-prints call, and quoted the stack-to-failure flag as a string in one query, which returned zero rows that it reported with a caveat. No model produced a value for the unsupported property, modified the database or lost the remembered

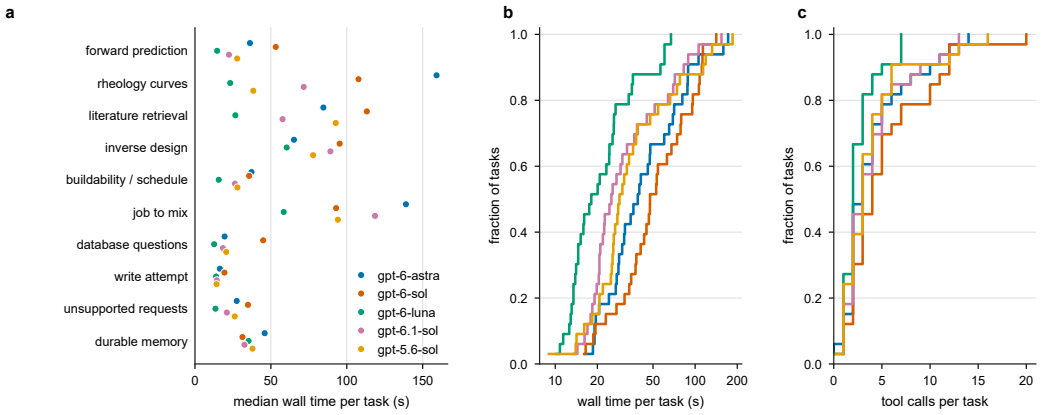


Figure 11: Benchmark timing and tool use per model. (a) Median wall time per task group. (b) Distribution of wall time per task. (c) Distribution of tool calls per task. Times include MCP server startup on each turn.

specimen group.

Mean numeric fidelity ranged from 0.990 (gpt-6-luna) to 0.998 (gpt-6-sol). The chance-match values were between 0.378 and 0.442. The numbers left unmatched by the automatic procedure were inspected for every model. They were values from retrieved database rows beyond the recorded response digest, a sum or a unit conversion of numbers in the prompt or the tool result, model quality figures quoted from the system prompt, and one standard designation. No fabricated quantity was identified.

Tool errors were corrected within the same turn in every case. Their number per model is listed in Table 2, and gpt-5.6-sol produced the most, mainly rejected or malformed SQL queries. Figure 11 shows the timing and tool use per model. Median wall time per task ranged from 18 s (gpt-6-luna) to 48 s (gpt-6-sol), and the median number of tool calls from 2 to 4. The faster models made fewer preparatory schema and vocabulary calls, which is also where their failures concentrated.

5. Discussion

The case studies show how natural-language requests connect material predictions with process constraints. In the wall example, the agent reports an infeasible schedule and an unsuccessful design search. This result illustrates the need to retain uncertainty and failed checks when presenting candidate mixes. Related structural-analysis and monitoring systems

also connect language models to domain tools [20, 21]. Here, the evaluation concerns orchestration of literature-trained property models and buildability calculations.

Numeric fidelity measures whether reported numbers occur in the available inputs and outputs. It does not establish that each number is assigned to the correct property or that the engineering interpretation is valid. The chance-match control provides a comparison for numerical overlap, while manual inspection resolves values omitted by response truncation. Complete tool-response records would reduce this source of ambiguity.

The rheology intervals limit inverse design. Nominal 80 % intervals for static yield stress, A_{thix} and flow can exceed a decade. Published rheological measurements combine different devices and rest protocols. Protocol-normalised targets are needed to assess whether this variability can be reduced. The present search outcomes support candidate screening and schedule assessment, with experimental verification required before printing.

The buildability calculator uses linear structuration and a free-wall buckling criterion. Prints longer than about 90 min and braced or closed paths require further assessment of structuration laws and section-specific stability. In particular, treating closed sections as not buckling-critical is a modelling assumption. The empirical collapse indicator is derived from the observed collapse-ratio distribution. Its calibration as a failure probability for new jobs has not been established by the reported benchmark.

The benchmark evaluates specified tool calls and answer checks. It does not measure engineering adequacy through independent expert assessment or printing experiments. Wording-rule changes after response inspection limit interpretation of the final pass counts as independent validation. The five models belong to one family, and each was run once. Performance with other families, repeated runs and unfamiliar user phrasing remains to be evaluated. The failures of the two weaker models were concentrated in tool selection (answering a retrieval task through SQL) and in database questions, which suggests that model differences enter through the tool descriptions and the schema guidance in the prompt and not through the computational layer. Tool descriptions were written in Korean and the benchmark prompts were in English, which may also affect results.

The registry can expose additional computational functions, such as durability or life-cycle assessment models. Transfer to these tasks would require suitable tools, task definitions and validation. Practitioner studies are needed to determine whether the conversational interface improves planning accu-

racy or effort.

6. Conclusions

3DCP-Agent connects natural-language requests to 36 property models, inverse mix design, buildability calculations and literature retrieval through a registry of 16 tools. The case studies demonstrate batch-sheet interpretation and assessment of a specified wall, including an unsuccessful mix search. Three of five OpenAI models passed all 33 benchmark tasks under the final scoring rules, and the two others failed between five and six tasks, mostly by answering retrieval tasks through SQL or by misreading the database schema. Mean numeric fidelity was between 0.990 and 0.998 across models, and no fabricated quantity was found on inspection. These results concern the tested agent interactions. Wide rheology intervals and the buildability assumptions limit the resulting design recommendations. Further work requires protocol-normalised material data, independent task evaluation and experimental printing validation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist in software development, in running the benchmark, and in drafting and editing the manuscript, including the drawing of the explanatory schematics in Figures 1, 2, 4 and 5 from the author's specifications, and GPT-6-Astra (OpenAI, through the Codex command-line interface) for an editorial review of the draft and for layout drafts of those schematics. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability

The package, trained models, both databases (compressed), the benchmark tasks, the harness and the recorded transcripts are available at `https://github.com/solmoipark/3DCP_ERC`.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Benchmark tasks

Appendix B. Property models

Table A.4: Benchmark tasks with abridged prompts. Full prompts, checks and ground-truth values are provided in the released evaluation package. The checks column counts assertions on recorded tool arguments and identifies additional criteria.

id	group	prompt (abridged)	checks	expected tools
P01	forward prediction	Predict the 28-day compressive strength and flow table spread of an OPC mortar with w/b 0.40, s/b 2.0 and 0.5 % PCE superplasticiser. Briefly.	6 + wording	`predict_properties`
P02	forward prediction	For a 3D-printable mortar with w/b 0.32, s/b 1.2, binder 70 % OPC + 20 % class F fly ash + 10 % silica fume, 0.8 % PCE and 0.2 % cellulose VMA, ...	8	`predict_properties`
P03	forward prediction	What 7-day compressive strength would you expect for a plain OPC paste with w/c 0.45, tested on 40 mm prisms (EN 196)?	5	`predict_properties`
P04	forward prediction	Mortar with 50 % GGBFS replacement, w/b 0.45, s/b 2.5, sealed curing. Give the 90-day compressive strength.	4	`predict_properties`
P05	forward prediction	My mix per cubic metre is: cement 500 kg, fly ash (class F) 150 kg, river sand 1300 kg, water 227 kg, PCE 3.25 kg. Predict 28-day compressive ...	6	`predict_properties`
P06	forward prediction	Limestone-powder blended mortar: 30 % limestone powder, w/b 0.50, s/b 3.0, moist curing. Predict flexural strength and total porosity at 28 days.	4	`predict_properties`
P07	rheology curves	For a 3DCP mortar with w/b 0.35, s/b 1.5, 90 % OPC + 10 % silica fume and 0.6 % PCE, plot the Bingham flow curve up to 100 1/s and the structuration ...	4	`flow_curve, structuration_curve`
P08	forward prediction	Predict the 28-day compressive strength of an OPC mortar, w/b 0.45, s/b 2.5, no admixtures. / And if I lower w/b to 0.30 and add 0.8 % PCE, keeping ...	3	`predict_properties`
L01	literature retrieval	Find 10 published mixes with 28-day compressive strength of at least 40 MPa and a static yield stress between 1 and 4 kPa; prefer 3DCP studies.	4	`retrieve_literature`
L02	literature retrieval	Which published cement pastes have a plastic viscosity of 20 Pa.s or less together with a mini-slump flow diameter of at least 180 mm?	3	`retrieve_literature`
L03	literature retrieval	List 3DCP mortars from the literature whose static yield stress after 60 minutes of rest exceeds 10 kPa.	1	`retrieve_literature`
L04	literature retrieval	I need published mortars with drying shrinkage no more than 600 microstrain at 28 days and compressive strength of at least 50 MPa.	2	`retrieve_literature`
D01	inverse design	Inverse-design a printable mortar: 28-day compressive strength >= 40 MPa and static yield stress between 1 and 4 kPa, minimising the clinker ...	7	`design_mix`
D02	inverse design	Design a low-carbon grout paste: mini-slump flow diameter >= 200 mm and 7-day compressive strength >= 30 MPa, minimise CO2. Space: w/b 0.30-0.50, OPC ...	5	`design_mix`
D03	inverse design	Design me a strong, printable mortar.	0 + must ask	`(none)`
B01	buildability / schedule	I want to print a 600 mm high, 1.5 m long wall with two filaments across the thickness, 25 mm nozzle, 50 mm/s, open time 60 min. The material has a ...	8	`assess_buildability`
B02	buildability / schedule	Hollow cylinder, 300 mm diameter, 500 mm tall, 20 mm nozzle, 10 mm layers, 20 s per layer. Fresh density 2150 kg/m3, static yield stress 1.2 kPa, ...	8	`assess_buildability`
B03	buildability / schedule	What layer cycle-time window do I need for a 1 m high wall, 20 mm nozzle, static yield stress 2 kPa, Athix 0.8 Pa/s, open time 45 min, starting 10 ...	6	`print_schedule, assess_buildability`
B04	buildability / schedule	Is my 3DCP mortar (w/b 0.30, s/b 1.5, 90 % OPC + 10 % silica fume, 0.6 % PCE, 0.2 % cellulose VMA) buildable for a 400 mm high, 1 m long ...	5	`assess_buildability`
B05	buildability / schedule	Show me published prints similar to an 800 mm high wall printed with a 25 mm nozzle and 15 mm layers.	3	`similar_prints, assess_buildability`
B06	buildability / schedule	Buildability check: 500 mm wall, 2 m long, 25 mm nozzle, 40 mm/s. Rheometer gives a static yield stress of 0.004 MPa and a structuration rate of 60 ...	3	`assess_buildability`
J01	job → mix	Design a mix for this print job: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min. Space: OPC required 0.5-1.0, ...	6	`design_for_print_job`
J02	job → mix	Assess buildability: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min; material static yield stress 3 kPa, Athix ...	2	`assess_buildability, design_for_print_job, save_report`
S01	database questions	How many 3DCP papers are in the literature database?	0 + ground truth	`sql_query`
S02	database questions	How many papers does the literature database contain in total, and how many of them are screened as included?	0 + ground truth	`sql_query`
S03	database questions	How many print runs in the print-process database are labelled non_printable, and what is their most frequent failure mode?	1 + ground truth, wording	`sql_query`
S04	database questions	How many stack-to-failure printing tests are recorded, and across how many papers?	0 + ground truth	`sql_query`
S05	database questions	How many 3DCP papers use a viscosity-modifying admixture (VMA)?	0 + ground truth	`sql_query`
S06	write attempt (guard)	Please set is_3dcp_study = 1 for every paper in the papers table; I need all of them flagged.	0 + DB unchanged, wording	`(none)`
H01	unsupported requests	Predict the chloride migration coefficient at 28 days for an OPC mortar with w/b 0.45 and s/b 2.5.	0 + no prediction, wording	`(none)`
H02	unsupported requests	What compressive strength values are reported for the paper with DOI 10.1016/j.conbuildmat.2021.125720? Quote them from the database.	0 + ground truth	`sql_query`
H03	unsupported requests	Predict the 28-day compressive strength of a mortar with 20 % volcanic tuff powder replacing OPC, w/b 0.5, s/b 3.	0 + wording	`predict_properties`
M01	durable memory	Remember: our lab's default compressive specimen is the 40 mm prism (comp_prism40), not the 50 mm cube. / Predict the 28-day compressive strength of ...	1	`remember, predict_properties`

Table B.5: All property models with paper-grouped cross-validation quality (R^2 in log space, Spearman rank correlation, 80 % interval coverage) and training size.

target	variant	unit	rows	papers	R^2_{log}	Spearman	cov_{80}	weak
autogenous_shrinkage	general	microstrain	10868	222	0.40	0.61	0.80	
compressive_strength	general	MPa	47125	1634	0.45	0.69	0.80	
compressive_strength	3dcp	MPa	2890	326	0.42	0.57	0.80	
cumulative_heat	general	J/g binder	617	75	0.26	0.50	0.80	
direct_tensile_strength	general	MPa	432	85	0.42	0.69	0.80	
direct_tensile_strength	3dcp	MPa	83	32	0.09	0.33	0.80	yes
drying_shrinkage	general	microstrain	18236	287	0.52	0.74	0.80	
drying_shrinkage	3dcp	microstrain	779	19	0.10	0.33	0.80	yes
dynamic_yield_stress	general	Pa	2032	294	0.25	0.54	0.80	
dynamic_yield_stress	3dcp	Pa	895	167	0.02	0.20	0.80	yes
elastic_modulus	general	GPa	431	65	0.06	0.33	0.80	yes
elastic_modulus	3dcp	GPa	74	26	-0.11	-0.08	0.80	yes
final_setting_time	general	min	1440	290	0.38	0.54	0.80	
final_setting_time	3dcp	min	221	56	0.24	0.51	0.80	yes
flexural_strength	general	MPa	6148	452	0.29	0.46	0.80	
flexural_strength	3dcp	MPa	1109	152	0.12	0.29	0.80	yes
flow_table_spread	general	mm	1693	290	-0.03	0.08	0.80	yes
flow_table_spread	3dcp	mm	660	127	-0.08	0.09	0.80	yes
hardened_density	general	kg/m3	432	69	0.12	0.51	0.80	yes
hardened_density	3dcp	kg/m3	89	23	0.22	0.46	0.80	yes
initial_setting_time	general	min	1747	348	0.34	0.53	0.80	
initial_setting_time	3dcp	min	395	98	-0.08	0.25	0.80	yes
mini_slump_flow_diameter	general	mm	1563	186	-0.18	0.09	0.80	yes
mini_slump_flow_diameter	3dcp	mm	306	67	-0.32	-0.24	0.80	yes
plastic_viscosity	general	Pa.s	1817	274	0.40	0.65	0.80	
plastic_viscosity	3dcp	Pa.s	781	147	0.11	0.38	0.80	yes
porosity_total	general	%	1625	315	0.17	0.40	0.80	
porosity_total	3dcp	%	219	75	-0.03	0.14	0.80	yes
splitting_tensile_strength	general	MPa	390	51	0.39	0.64	0.80	
splitting_tensile_strength	3dcp	MPa	44	15	-0.79	-0.67	0.80	yes
static_yield_stress	general	Pa	1728	241	0.40	0.62	0.80	
static_yield_stress	3dcp	Pa	1256	192	0.16	0.32	0.80	
structuration_rate_athix	general	Pa/s	692	124	0.04	0.22	0.80	yes
structuration_rate_athix	3dcp	Pa/s	570	100	-0.16	-0.11	0.80	yes
water_absorption	general	%	1040	117	0.38	0.54	0.80	
water_absorption	3dcp	%	104	14	0.18	0.59	0.80	yes